QUESTIONS 31 / 31 completed

Arithmetic

ARITHMETIC

Write a Java program that takes two integers as input from the user and calculates their sum, difference, product, and quotient using arithmetic operators.

PROGRAM EDITOR

Java | Run | Save

```java
import java.util.Scanner;
class Main{
  public static void main(String Args[])
  {
    int a,b;
    Scanner sc= new Scanner(System.in);
    System.out.println("enter two numbers:");
    a= sc.nextInt();
    b= sc.nextInt();
    System.out.println("addition: "+(a+b)+"\n"+"subtraction: "+(a-b));
    System.out.println("multiplication: "+(a*b)+"\n"+"division: "+(a/b));

  }
}
```

INPUT

```
10
5
```

OUTPUT

```
enter two numbers:
addition: 15
subtraction: 5
multiplication: 50
division: 2
```

QUESTIONS 31 / 31 completed

Search

SEARCH

Write a Java program to implement binary search on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java | Run | Save

```java
import java.util.Arrays;
class binarysearch{
  public static int binarysearch(int arr[],int target){
    int low=0,high=arr.length-1;
    while(low<=high){
      int mid=low+(high-low)/2;
      if(arr[mid]==target){
        return mid;
      }
      else if(arr[mid]<target){
        low=mid+1;
      }
      else{
        high=mid-1;
      }
    }
    return -1;
  }
  public static void main(String arg[]){
    int arr[]={2,5,1,9,10,21,56,34,42,12};
    int target=56;
    Arrays.sort(arr);
    int result=binarysearch(arr,target);
    if(result==-1){
      System.out.println("not found");

    }
    else{
      System.out.println("the number found at index : "+result);
    }
  }
}
```

OUTPUT

```
the number found at index : 9
```

QUESTIONS 31 / 31 completed

Factorial

FACTORIAL

Write a Java program to calculate the factorial of a given number using loops.

PROGRAM EDITOR

Java ✓ Updated Run Save

```
import java.util.Scanner;
class N{
  public static void main(String Args[]){
    int a=1,i,n;
    Scanner sc= new Scanner(System.in);
    n=sc.nextInt();
    for(i=1;i<=n;i++){
     a*=i;
    }
     System.out.println("the factorial of "+n+" is "+a);
  }
}
```

INPUT

5

OUTPUT

```
the factorial of 5 is 120
```

QUESTIONS 31 / 31 completed

Fibonacci

FIBONACCI

Write a Java program to print the Fibonacci series up to a given number using loops.

PROGRAM EDITOR

Java Run Save

```
import java.util.Scanner;
class Main{
  public static void main(String Arg[]){
    int a=0,b=1,c,i;
    Scanner f= new Scanner(System.in);
    int n=f.nextInt();
    System.out.println("Fibonacci series\n");
    for(i=0;i<=n;i++){
      System.out.print(a+" ");
      c=a+b;
      a=b;
      b=c;
    }
  }
}
```

INPUT

10

OUTPUT

```
Fibonacci series

0 1 1 2 3 5 8 13 21 34 55
```

QUESTIONS 31 / 31 completed

GCD

GCD

Write a Java program to find the GCD (Greatest Common Divisor) of two numbers using loops and conditional statements.

PROGRAM EDITOR

Java Run Save

```
import java.util.Scanner;
class Main{
  public static void main(String args[]){
    Scanner sc=new Scanner(System.in);
    System.out.println("enter the two numbers to find the gcd :");
    int a=sc.nextInt();
    int b=sc.nextInt();
    while(b!=0){
      int temp=b;
      b=a%b;
      a=temp;
    }
    System.out.println("The GCD is "+a);
  }
}
```

INPUT

6
9

OUTPUT

```
enter the two numbers to find the gcd :
The GCD is 3
```

QUESTIONS 31 / 31 completed

Leap

LEAP

Write a Java program to check whether a given year is a leap year or not using conditional statements.

PROGRAM EDITOR

Java Run Save

```java
import java.util.Scanner;
class A{
  public static void main(String Args[])
  {
    int a;
    Scanner sc= new Scanner(System.in);
    System.out.println("enter the year to check if it is leap or not:");
    a=sc.nextInt();
    if ((a%400==0)||((a%4==0)&& (a%100!=0)))
      System.out.println("the given year is a leap year");
    else
      System.out.println("the given year is not a leap year");
  }
}
```

INPUT

2024

OUTPUT

```
enter the year to check if it is leap or not:
the given year is a leap year
```

QUESTIONS 31 / 31 completed

MAX

MAX

Write a Java program to find the maximum of three numbers using conditional statements.

PROGRAM EDITOR

Java Run Save

```java
import java.util.Scanner;
class Main{
  public static void main(String Args[]){
    int a,b,c;
    Scanner sc= new Scanner(System.in);
    System.out.println("enter three numbers to find the maximum of three numbers:");
    a=sc.nextInt();
    b=sc.nextInt();
    c=sc.nextInt();
    if(a>b && b>c)
      System.out.println(a+" is the maximum of the three numbers");
    else if(b>a && b>c)
      System.out.println(b+" is the maximum of the three numbers");
    else
      System.out.println(c+" is the maximum of all three numbers");
  }
}
```

INPUT

5
4
8

OUTPUT

```
enter three numbers to find the maximum of three numbers:
8 is the maximum of all three numbers
```

QUESTIONS 31 / 31 completed

Prime

PRIME

Write a Java program to check whether a given number is a prime number or not using loops and conditional statements.

PROGRAM EDITOR

Java | Run | Save

```
import java.util.Scanner;
class prime{
  public static void main(String args[]){
    Scanner sc=new Scanner(System.in);
    int n=0;
    System.out.println("Enter the number to check whether the given number is prime or n
    int a=sc.nextInt();
    for(int i=1;i<=a;i++){
      if(a%i==0)
        n=n+1;
    }
    if(n==2)
      System.out.println("The given number is prime");
    else
      System.out.println("The given number is not a prime number");
  }
}
```

INPUT

11

OUTPUT

```
Enter the number to check whether the given number is prime or not:
The given number is prime
```

RT-Pattern

RT-PATTERN

Write a Java program to print the following pattern using loops:*

**

***

****

*****

```
import java.util.Scanner;
class Main{
  public static void main(String Args[]){
    int i,j,n;
    Scanner N= new Scanner(System.in);
    System.out.println("enter the no. of rows");
    n=N.nextInt();
    for(i=1;i<=n;i++){
      for(j=1;j<=i;j++){
        System.out.print("*");
      }
      System.out.println();
    }
  }
}
```

5

OUTPUT

```
enter the no. of rows
*
**
***
****
*****
```

QUESTIONS 31 / 31 completed

Sort

SORT

Write a Java program to implement selection sort on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java | Run | Save

```
import java.util.Scanner;
import java.util.Arrays;
class Selsort{
  public static void main(String args[]){
    Scanner sc=new Scanner(System.in);
    System.out.println("how many elements do you want to sort: ");
    int n=sc.nextInt();
    int sort[]=new int[n];
    for(int k=0;k<n-1;k++){
      sort[k]=sc.nextInt();
    }
    for(int i=0;i<sort.length-1;i++){
      int min=i;
      for(int j=i+1;j<sort.length;j++){
        if(sort[j]<sort[min]){
          min=j;}
          int temp=sort[min];
          sort[min]=sort[i];
          sort[i]=temp;      }      }
    System.out.println("the sorted array: "+Arrays.toString(sort));
  }     }
```

INPUT

5
7
1
6
2
3

OUTPUT

```
how many elements do you want to sort:
the sorted array: [0, 1, 2, 6, 7]
```

QUESTIONS 31 / 31 completed

Table

TABLE

Write a Java program to print the multiplication table of a given number using loops.

PROGRAM EDITOR

Java Run Save

```java
import java.util.Scanner;
class table{
  public static void main(String args[]){
    Scanner M= new Scanner(System.in);
    System.out.println("Enter what multiplication table should be displayed:");
    int a=M.nextInt();
    for(int i=1;i<=10;i++){
      System.out.println(a+" x "+i+" = "+(a*i));
    }
  }
}
```

INPUT

32

OUTPUT

```
Enter what multiplication table should be displayed:
32 x 1 = 32
32 x 2 = 64
32 x 3 = 96
32 x 4 = 128
32 x 5 = 160
32 x 6 = 192
32 x 7 = 224
32 x 8 = 256
32 x 9 = 288
32 x 10 = 320
```

QUESTIONS 31 / 31 completed

Factorial

FACTORIAL

2. Write a Java program to find the factorial of a given number using recursion.

PROGRAM EDITOR

Java Run Save

```java
import java.util.Scanner;
class Main{
  public static void main(String args[]){
    Scanner sc=new Scanner(System.in);
    int n=sc.nextInt();
    int a=1;
    while(n!=0){
      a=a*n;
      n=n-1;
    }
    System.out.println("The factorial of the given number is "+a);
  }
}
```

INPUT

4

OUTPUT

```
The factorial of the given number is 24
```

QUESTIONS 31 / 31 completed

LT-Pattern

LT-PATTERN

Write a Java program to print the following pattern using loops:*****
****
***
**
*

PROGRAM EDITOR   Java   Run   Save

```java
import java.util.Scanner;
class pattern{
  public static void main(String Args[]){
    Scanner sc= new Scanner(System.in);
    System.out.println("enter the number of rows");
    int r=sc.nextInt();
    for(int i=r;i>=1;i--){
      for(int j=1;j<=i;j++){
        System.out.print("*");
      }
      System.out.println();
    }
  }
}
```

INPUT

5

OUTPUT

```
enter the number of rows
*****
****
***
**
*
```

QUESTIONS 31 / 31 completed

LCM

LCM

Write a Java program to find the LCM (Least Common Multiple) of two numbers using loops and conditional statements.

PROGRAM EDITOR   Java   Run   Save

```java
import java.util.Scanner;
class LCM{
  public static void main(String args[]){
    Scanner sc=new Scanner(System.in);
    System.out.println("enter 2 numbers to find LCM : ");
    int a=sc.nextInt();
    int b=sc.nextInt();
    int lcm=lcm(a,b);
    System.out.println("The LCM = "+lcm);
  }
  static int gcd(int i,int j){
    while(j!=0){
      int t=j;
      j=i%j;
      i=t;
    }
    return i;
  }
  static int lcm(int a, int b){
    return (a*b)/gcd(a,b);
  } }
```

INPUT

2
5

OUTPUT

```
enter 2 numbers to find LCM :
The LCM = 10
```

QUESTIONS 31 / 31 completed

D to B

D TO B

Write a Java program to convert a decimal number to binary using loops and conditional statements.

PROGRAM EDITOR   Java   Run   Save

```java
import java.util.Scanner;
class dtob{
  public static void main(String args[]){
    Scanner sc= new Scanner(System.in);
    System.out.println("enter the decimal number: ");
    int a=sc.nextInt();
    int b;
    String binary="";
    if(a==0){
      binary="0";
    }
    while(a>0){
      b=a%2;
      binary=b+binary;
      a=a/2;
    }
    System.out.println("The binary value of given decimal number is "+binary);
  }
}
```

INPUT

10

OUTPUT

```
enter the decimal number:
The binary value of given decimal number is 1010
```

QUESTIONS 31 / 31 completed

B to D

B TO D

Write a Java program to convert a binary number to decimal using loops and conditional statements.

PROGRAM EDITOR

Java | Run | Save

```
import java.util.Scanner;
class bin{
  public static void main(String Args[]){
    int d=0;
    int i=0;
    Scanner sc= new Scanner(System.in);
    System.out.println("Enter the binary number: ");
    int a=sc.nextInt();
    while(a>0){
      int b=a%10;
      if(b==0||b==1){
      d=d+b*(int)(Math.pow(2,i));
      i+=1;
      a=a/10;       }
      else{
        System.out.println("invalid binary number");
        return;     }
    }
    System.out.println("Decimal value of the given binary number is:"+d);
  }
}
```

INPUT

1101

OUTPUT

```
Enter the binary number:
Decimal value of the given binary number is:13
```

QUESTIONS 31 / 31 completed

Sum

SUM

Write a Java program to calculate the sum of the digits of a given number using loops and conditional statements.

PROGRAM EDITOR

Java | Run | Save

```
import java.util.Scanner;
class sum{
  public static void main(String Args[]){
    int s=0;
    Scanner sc= new Scanner(System.in);
    System.out.println("Enter the digits:");
    int a=sc.nextInt();
    if(a<0){
      a=-a;
    }
    while(a>0){
      int x=a%10;
      a/=10;
      s+=x;

    }
    System.out.println("the sum of the digits="+s);

  }
}
```

INPUT

-1234567890

OUTPUT

```
Enter the digits:
the sum of the digits=45
```

QUESTIONS 31 / 31 completed

Product

PRODUCT

Write a Java program to calculate the product of the digits of a given number using loops and conditional statements.

PROGRAM EDITOR

Java | Run | Save

```
import java.util.Scanner;
class pro{
  public static void main(String Args[]){
    int p=1;
    Scanner sc= new Scanner(System.in);
    System.out.println("enter the digits:");
    int a=sc.nextInt();
    if(a<0)
      a=-a;
    while(a>0){
      int x=a%10;
      a/=10;
      p*=x;

    }
    System.out.println("the product of the digits:"+p);
  }
}
```

INPUT

-2345

OUTPUT

```
enter the digits:
the product of the digits:120
```

QUESTIONS 31 / 31 completed

Power

POWER

Write a Java program to find the power of a number using loops and conditional statements.

PROGRAM EDITOR

Java Run Save

```
import java.util.Scanner;
class power{
 public static void main(String args[]){
   Scanner sc= new Scanner(System.in);
   int a,b;

   System.out.println("Enter the base and the power:");
   a=sc.nextInt();
   b=sc.nextInt();
   int c=1;
   if(b>=0){
   for(int i=1;i<=b;i++){
     c*=a;
   }
   }
   else{
     System.out.println("invalid power");
   }
   System.out.println(a +" to the power of "+ b +" is " + c);
 }
}
```

INPUT

```
5
0
```

OUTPUT

```
Enter the base and the power:
5 to the power of 0 is 1
```

QUESTIONS 31 / 31 completed

Area Circle

AREA CIRCLE

Write a Java program to calculate the area of a circle using the radius input by the user.

PROGRAM EDITOR

Java ✓ Updated Run Save

```
import java.util.Scanner;
class area{
 public static void main(String Args[]){
   Scanner sc= new Scanner(System.in);
   float r;
   System.out.println("enter the radius of the circle in meters:");
   r=sc.nextFloat();
   double area=(22.0/7)*r*r;
   System.out.println("the area of the circle is "+area+" sq.units");
 }
}
```

INPUT

```
5
```

OUTPUT

```
enter the radius of the circle in meters:
the area of the circle is 78.57142857142857 sq.units
```

QUESTIONS 31 / 31 completed

Shpere

SHPERE

Write a Java program to calculate the volume of a sphere using the radius input by the user.

PROGRAM EDITOR

Java Run Save

```
import java.util.Scanner;
class volume{
 public static void main(String Args[]){
   Scanner sc= new Scanner(System.in);
   System.out.println("Enter the radius in meters:");
   float r=sc.nextFloat();
   double v=(4.0/3)*(22.0/7)*r*r*r;
   System.out.println("the volume of the sphere is "+v+" cubic.units");
 }
}
```

INPUT

```
1
```

OUTPUT

```
Enter the radius in meters:
the volume of the sphere is 4.19047619047619 cubic.units
```

QUESTIONS 31 / 31 completed

Palindrome

PALINDROME

Write a Java program to check whether a given string is a palindrome or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run Save

```
import java.util.Scanner;
class Pal{
  public static void main(String args[]){
    Scanner sc= new Scanner(System.in);
    System.out.println("enter the string: ");
    String str=sc.next();
    String rev="";
    for(int i=str.length()-1;i>=0;i--){
      rev=rev+str.charAt(i);
    }
    System.out.println("The reversed string: "+rev);
    if(rev.equals(str)){
      System.out.println("The given string is palindrome");

    }
    else{
      System.out.println("The given string is not a palindrome");
    }
  }
}
```

INPUT

malayalam

OUTPUT

```
enter the string:
The reversed string: malayalam
The given string is palindrome
```

QUESTIONS 31 / 31 completed

Armstrong

ARMSTRONG

Write a Java program to check whether a given number is an Armstrong number or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run Save

```
import java.util.Scanner;
class Main{
  public static void main(String args[]){
    Scanner sc=new Scanner(System.in);
    int n=sc.nextInt();
    int a=0;
    int c=0;
    int b=n;
    while(b>0){
      a=b%10;
      c=c+(a*a*a);
      b=b/10;

    }
    if(n==c){
      System.out.println("the given number is armstrong");
    }
    else{
      System.out.println("the given number is not armstrong");
    }    }      }
```

INPUT

153

OUTPUT

```
the given number is armstrong
```

QUESTIONS 31 / 31 completed

RT-num.Pattern

RT-NUM.PATTERN

Write a Java program to print the following pattern using loops:1
22
333
4444
55555

PROGRAM EDITOR

```
class pat{
  public static void main(String args[]){
    int i,j;
    int k=1;
    for(i=1;i<=5;i++){
      for(j=1;j<=i;j++){
        System.out.print(k+" ");
      }
      k++;
      System.out.println();
    }

  }
}
```

OUTPUT

```
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
```

QUESTIONS 31 / 31 completed

Max Array

MAX ARRAY

Write a Java program to find the largest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

```java
import java.util.Arrays;
class arr{
  public static void main(String args[]){
    int arr[]= {1,5,3,9,7};
    int max=0;
    for(int i=0;i<arr.length;i++){
      if(max<arr[i])
        max=arr[i];
    }
    System.out.println("The largest number:"+max);
  }
}
```

OUTPUT

```
The largest number:9
```

QUESTIONS 31 / 31 completed

2nd Max

2ND MAX

Write a Java program to find the second largest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java Run Save

```java
import java.util.Scanner;
class main{
  public static void main(String args[]){
    Scanner sc= new Scanner(System.in);
    System.out.println("how many numbers do you want to enter:");
    int n= sc.nextInt();
    int a[]=new int[n];
    System.out.println("Enter the numbers");
    for(int i=0;i<a.length;i++){
      a[i]=sc.nextInt();
    }
    for(int j=0;j<a.length;j++){
      for(int k=0;k<a.length;k++){
        if(a[j]>a[k]){

          int x=a[j];
          a[j]=a[k];
          a[k]=x;
        }
      }
    }
    System.out.print("your numbers after sorted:");
    for(int l=0;l<a.length;l++){
      System.out.print(a[l]);
    }
    System.out.println("\n2nd largest:"+a[1]);
  }
}
```

INPUT

```
5
1
4
2
5
3
```

OUTPUT

```
how many numbers do you want to enter:
Enter the numbers
your numbers after sorted:54321
2nd largest:4
```

QUESTIONS 31 / 31 completed

Min

MIN

Write a Java program to find the smallest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java Run Save

```java
import java.util.Arrays;
import java.util.Scanner;
class min{
  public static void main(String args[]){
    int a[]=new int[5];
    int min;
    Scanner sc=new Scanner(System.in);
    System.out.println("Enter any 5 numbers to find the minimum:");
    System.out.print("Your numbers:");
    for(int i=0;i<5;i++){
      a[i]=sc.nextInt();
      System.out.print(a[i]);
    }
    min=a[0];
    for(int j=0;j<5;j++){
      if(min>a[j])
        min=a[j];
    }
    System.out.println("\nminimum number: "+min);
  }
}
```

INPUT

```
1
3
2
6
5
```

OUTPUT

```
Enter any 5 numbers to find the minimum:
Your numbers:13265
minimum number: 1
```

QUESTIONS 31 / 31 completed

2nd Min

2ND MIN

Write a Java program to find the second smallest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR Java Run Save

```java
import java.util.Arrays;
import java.util.Scanner;
class z{
  public static void main(String args[]){
    Scanner sc=new Scanner(System.in);
    System.out.println("how many elements do you want enter:");
    int n=sc.nextInt();
    int a[]=new int[n];
    System.out.println("enter the numbers");
    for(int i=0;i<a.length;i++){
      a[i]=sc.nextInt();
    }
    Arrays.sort(a);
    System.out.println("second smallest:"+a[1]);
  }
}
```

INPUT

```
7
12
23
21
52
53
42
46
```

OUTPUT

```
how many elements do you want enter:
enter the numbers
second smallest:21
```

QUESTIONS 31 / 31 completed

Sort-B

SORT-B

Write a Java program to implement bubble sort on an array of integers using loops and conditional statements.

PROGRAM EDITOR Java Run Save

```java
import java.util.Scanner;
import java.util.Arrays;
class Main{
  public static void main(String[] args){
    Scanner sc= new Scanner(System.in);
    System.out.println("how many elements do you want to sort");
    int n=sc.nextInt();
    int arr[]=new int[n];
    for(int k=0;k<n;k++){
      arr[k]=sc.nextInt();
    }
    for(int i=0;i<n-1;i++){
      boolean swapped=false;
      for(int j=0;j<n-1-i;j++){
        if(arr[j]>arr[j+1]){
          int temp=arr[j];
          arr[j]=arr[j+1];
          arr[j+1]=temp;
          swapped=true;
        }
      }
      if(swapped==false)
        break; }
      System.out.println("the sorted elemenys:"+Arrays.toString(arr));
    } }
```

INPUT

```
5
1
6
2
5
3
```

OUTPUT

```
how many elements do you want to sort
the sorted elemenys:[1, 2, 3, 5, 6]
```

Sort-I

SORT-I

Write a Java program to implement insertion sort on an array of integers using loops and conditional statements.

```
import java.util.Arrays;
class insertionsort{
  public static void main(String args[]){
    int arr[]={4,5,3,1,2};
    for(int i=1;i<arr.length;i++){
      int key=arr[i];
      int j=i-1;
      while(j>=0&&arr[j]>key){
        arr[j+1]=arr[j];
        j--;
      }
      arr[j+1]=key;
    }
    System.out.println("Sorted array: "+Arrays.toString(arr));
  }
}
```

OUTPUT

```
Sorted array: [1, 2, 3, 4, 5]
```